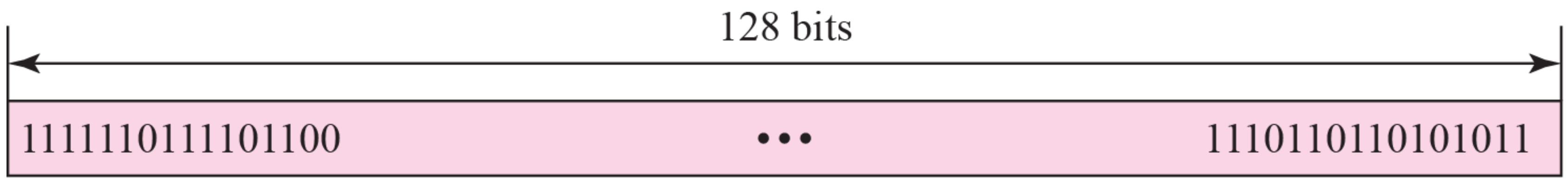
IPv6

IPv6 address formats

- IPv6 addresses are 128 bits long. IPv4 addresses are only 32 bits long
- IPv6 addresses are written as eight hexadecimal groups. Each hexadecimal group, separated by a colon (:), consists of a 16-bit hexadecimal value. The following is an example of the IPv6 format:
  - xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx
- A group of xxxx represents the 16-bit hexadecimal value. Each individual x represents a 4-bit hexadecimal value. The following is an example of a possible IPv6 address:
  - 2001:0340:0000:0000:0000:F673:0029:0564

IPv6 addresses

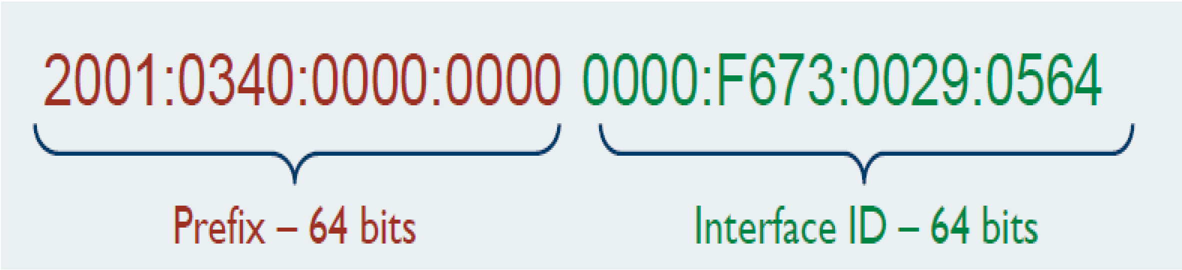


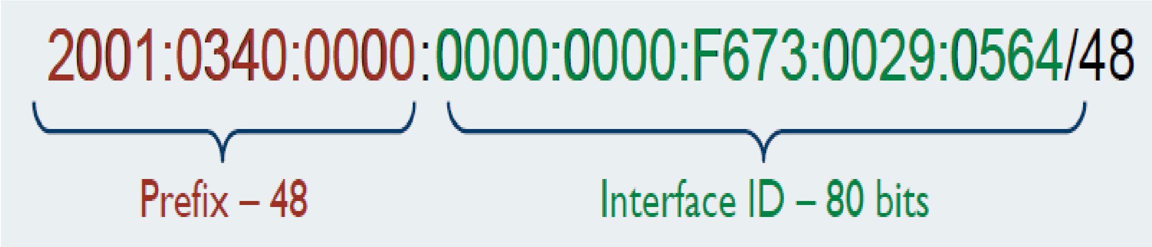
Colon hexadecimal notation

FDEC : BA98 : 7654 : 3210 : ADBF : BBFF : 2922 : FFFF

Prefix and interface ID

- IP addresses combine, in a single address, a network identifier (called the prefix) and a device identifier (the interface ID).
- The point at which to split the address into these two portions is given by the prefix length.
- The prefix length is written as /xx at the end of the address; e.g.
    - 2001:0340:0000:0000:0000:F673:0029:0564/64





Address optimization

- To make IPv6 addresses easier to write, some zeros can be removed. The leading zeros in a 4-digit block can be removed.
- Also, contiguous sets of 4 zeros, and their separating colons can be completely removed.

2001:0340:0000:0000:0000:F673:0029:0564

Address optimization Cont

- To avoid ambiguity, it is only possible to have one place in the address where a continuous set of 0s is replaced by ::

So 2001:0340:0000:0000:F673:0000:0000:0564

can be written as

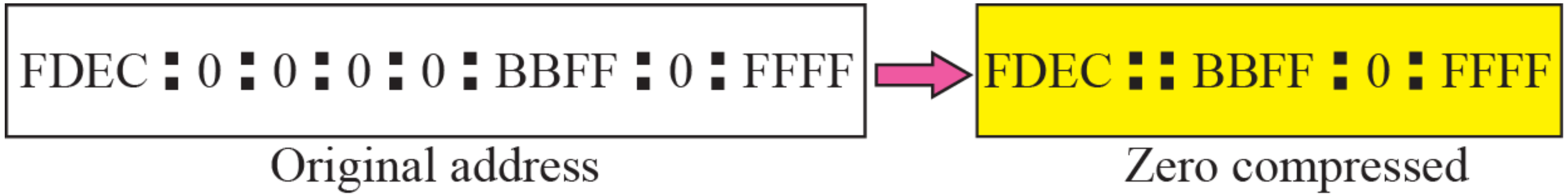
2001:340::F673:0000:0000:564

or

2001:340:0000:0000:F673::564

but NOT

2001:340::F673::564


FDEC : 0 : 0 : 0 : 0 : BBFF : 0 : FFFF
Original address
FDEC :: BBFF : 0 : FFFF
Zero compressed

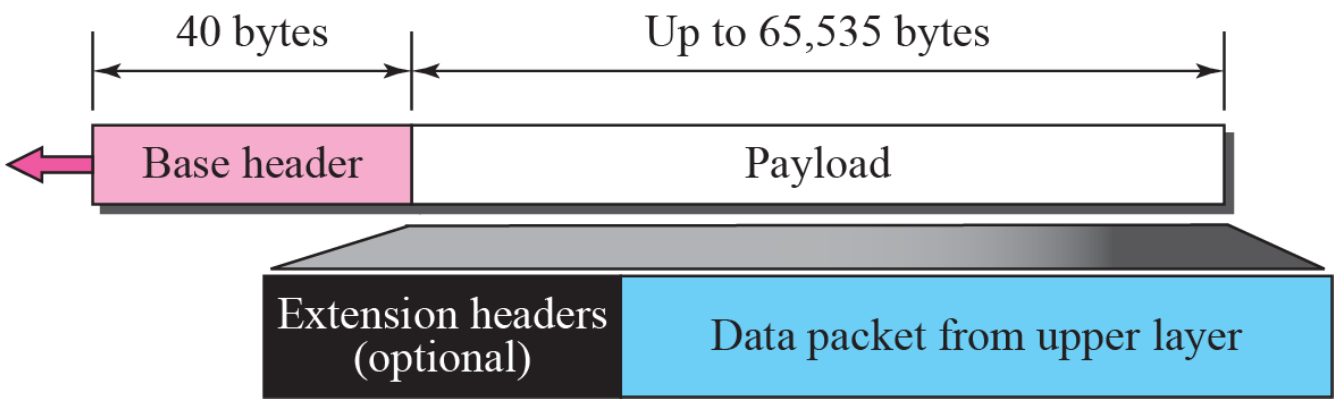
CIDR address

FDEC :: BBFF : 0 : FFFF/60

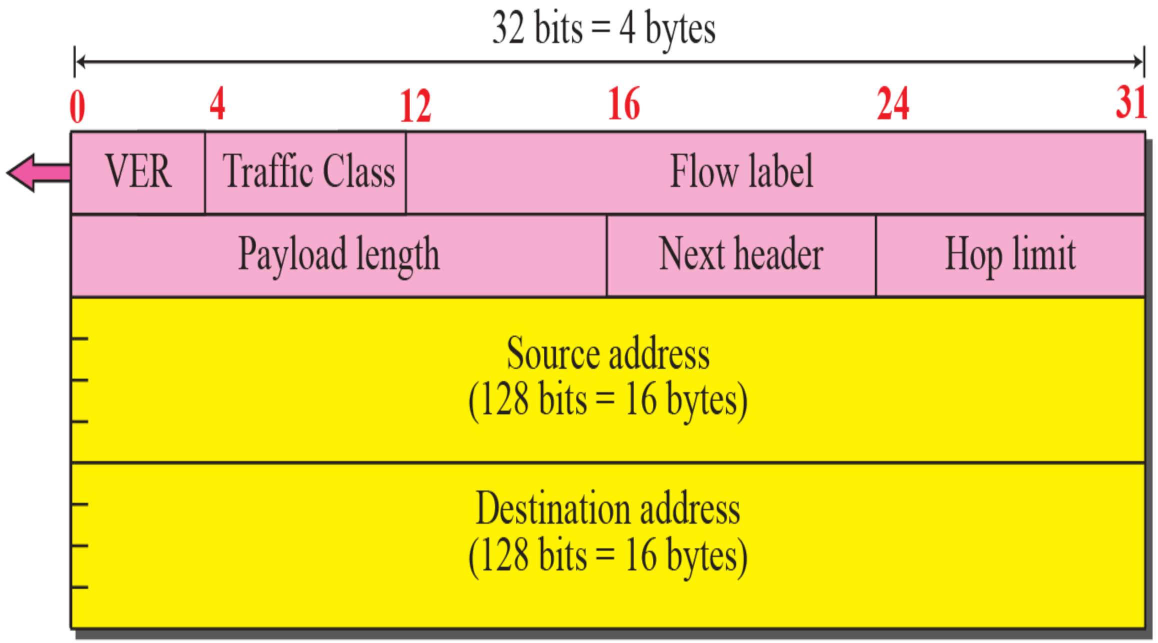
IPv6 Header Format

- Forty bytes long (IPv4 Header is 20 bytes)
- Each IPv6 address is four times the length of an IPv4 address
- The IPv6 header no longer contains the header length, identification, flags, fragment offset and header checksum fields
- Some of these options have been placed in extension headers
- The ‘Time To Live’ field has been replaced with a hop limit
- IPv4 Type of Service field is now replaced with a Traffic Class field

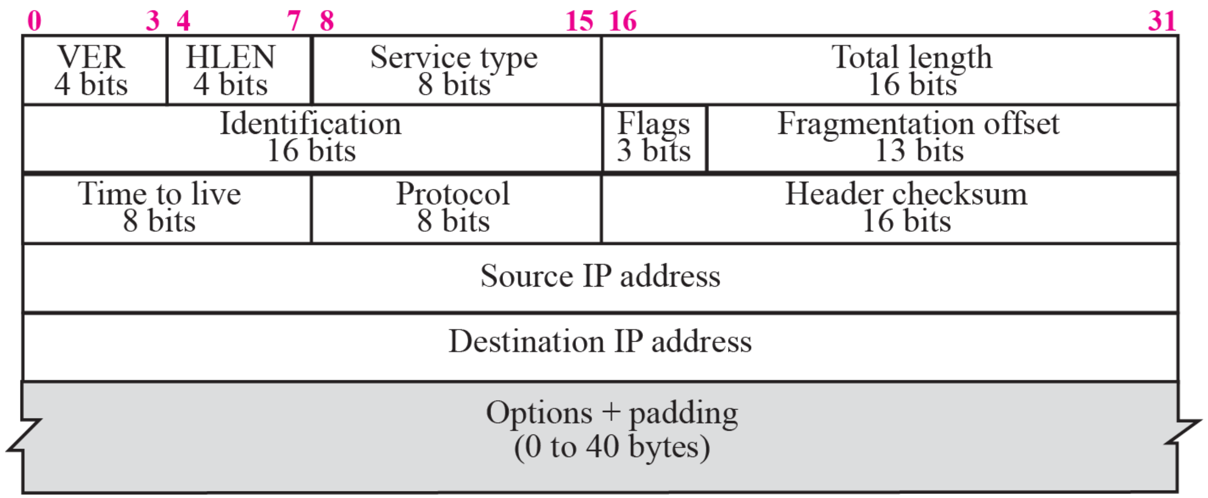
IPv6 datagram



Format of the base header



IPv6



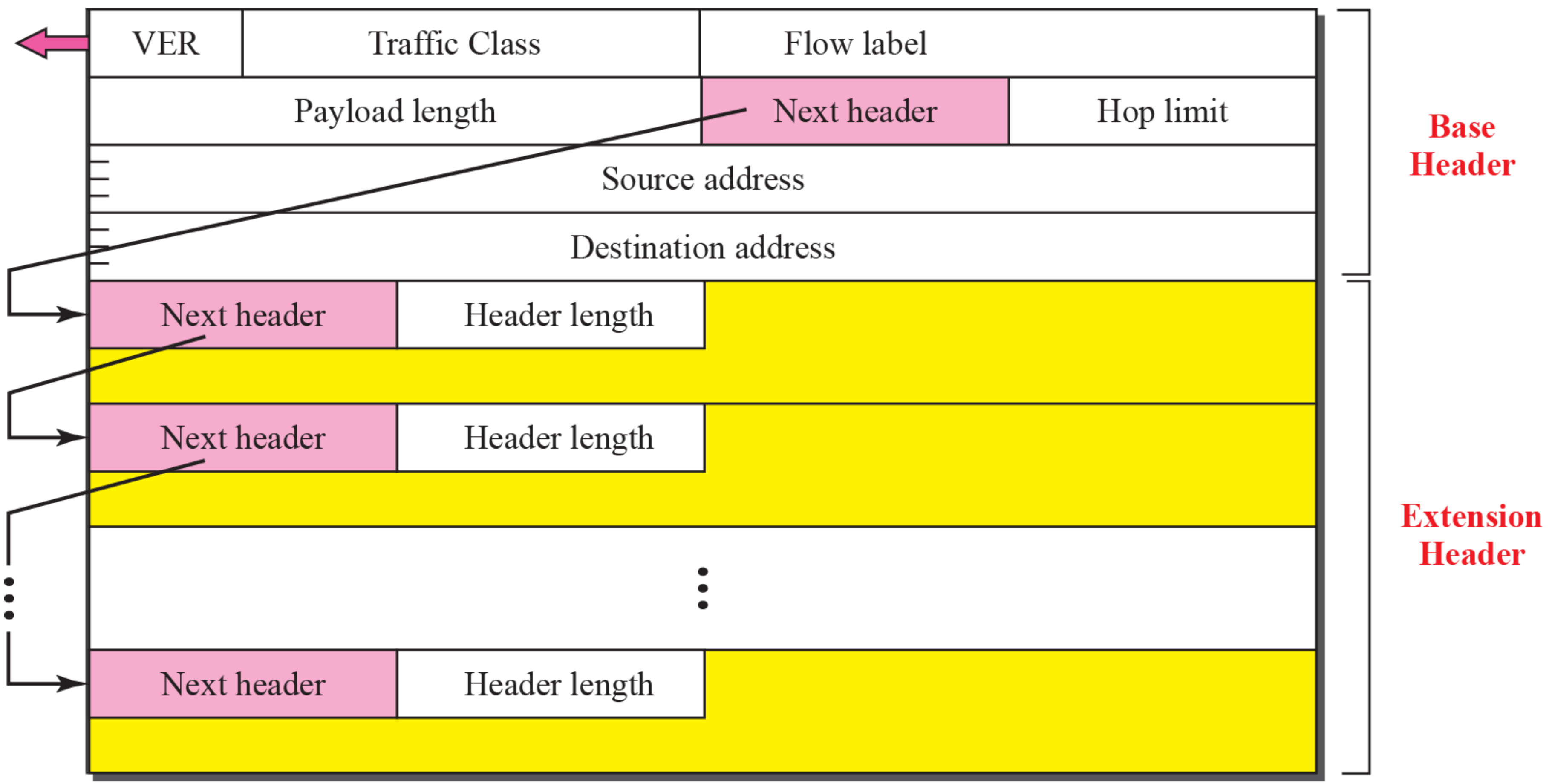
b. Header format

IPv4

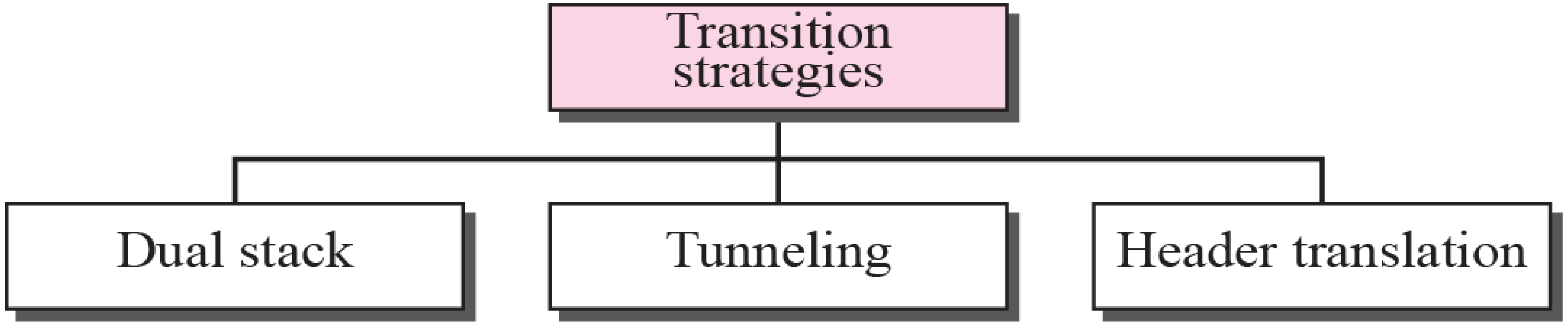
Table 27.1 *Next Header Codes*

Code	*Next Header*	*Code*	*Next Header*
0	Hop-by-hop option	44	Fragmentation
2	ICMP	50	Encrypted security payload
6	TCP	51	Authentication
17	UDP	59	Null (No next header)
43	Source routing	60	Destination option

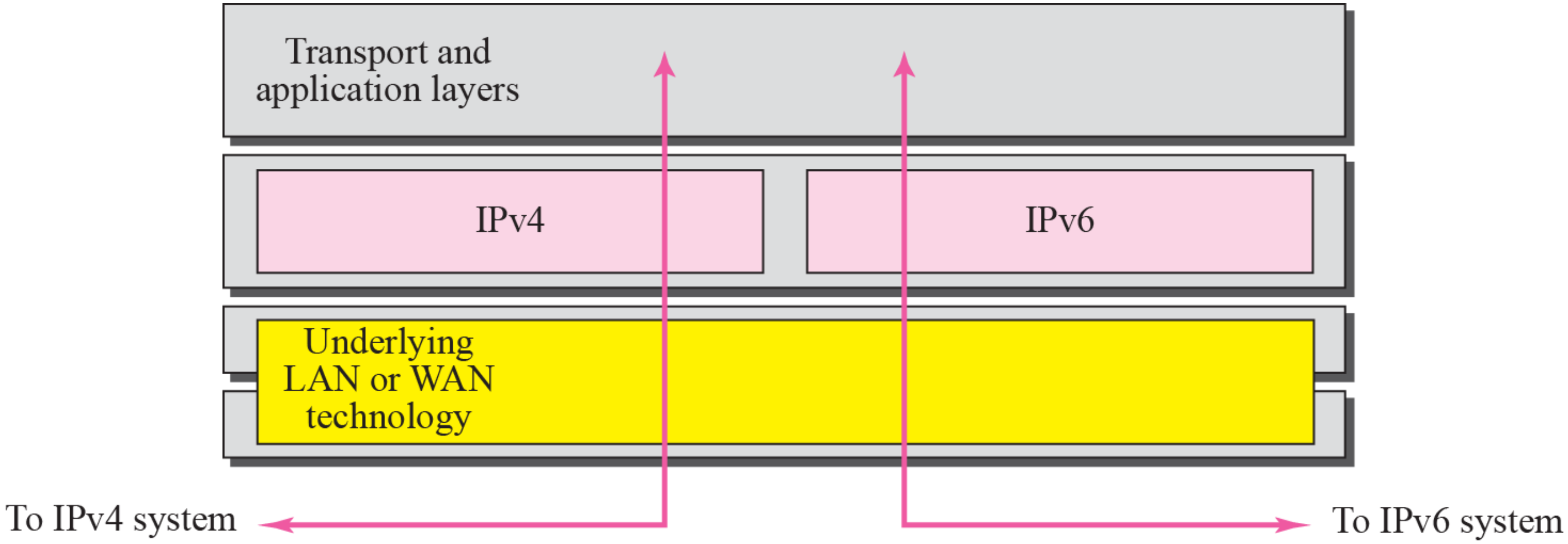
Extension header format



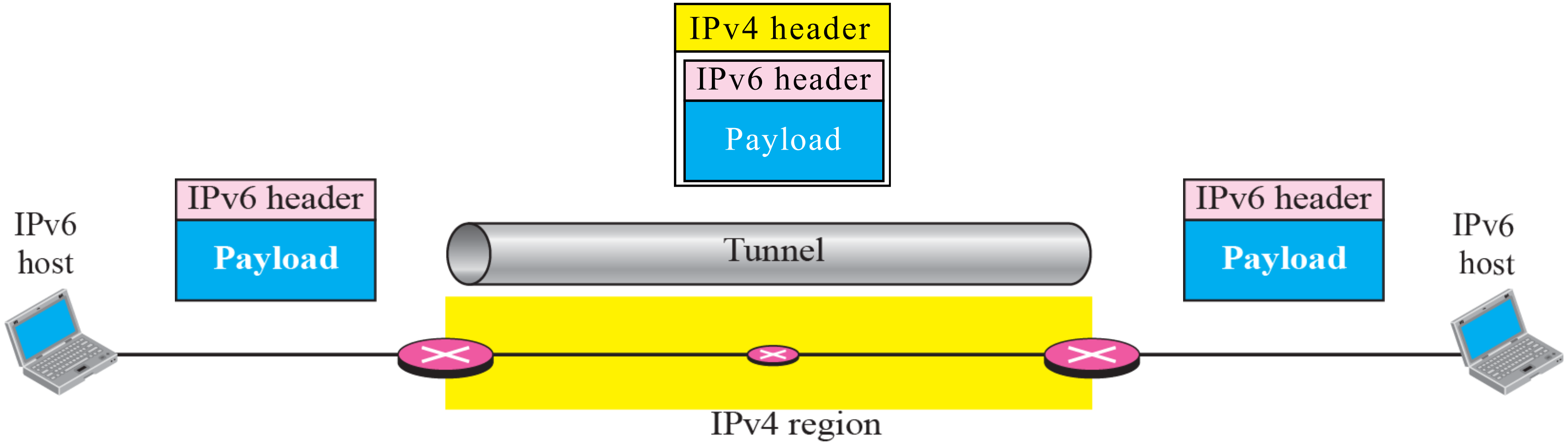
Three transition strategies



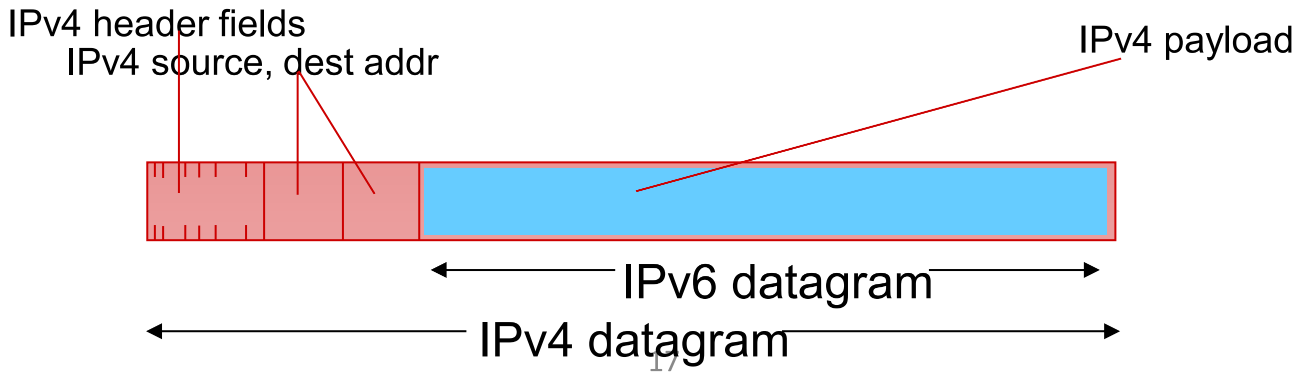
Dual stack



Device has both the stacks so can handle both ipv4 and ipv6 packets.

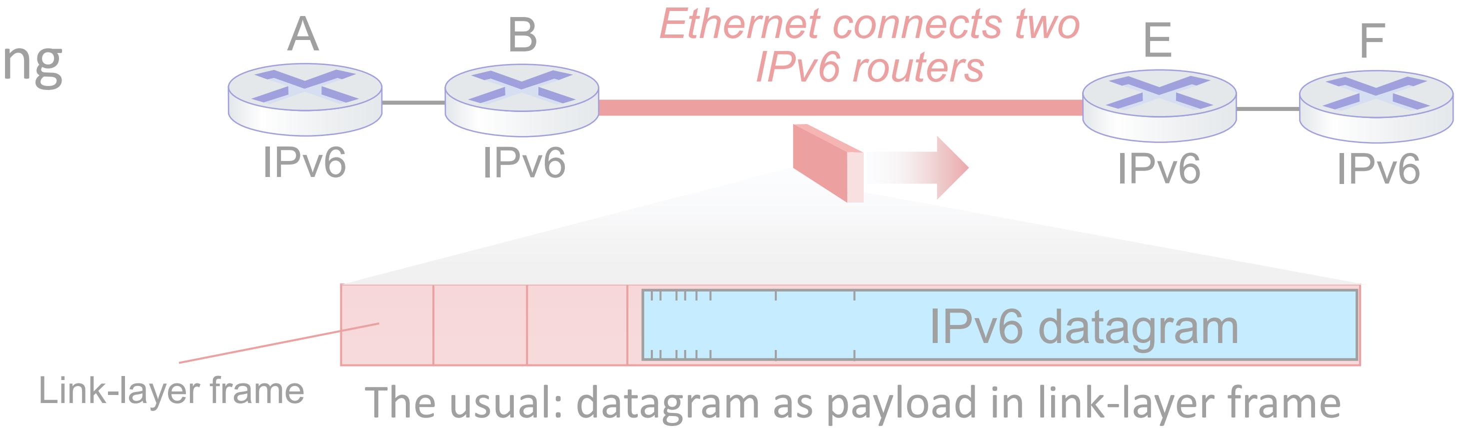


- tunneling: IPv6 datagram carried as *payload* in IPv4 datagram among IPv4 routers ("packet within a packet")
  - tunneling used extensively in other contexts (4G/5G)



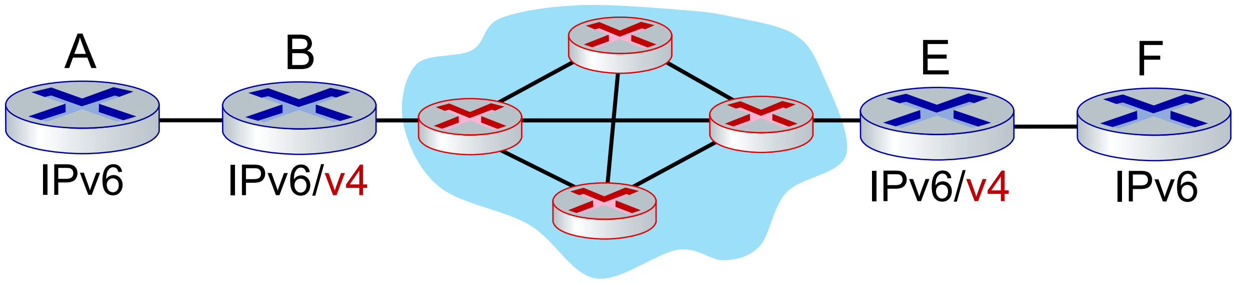
Tunneling and encapsulation

Ethernet connecting two IPv6 routers:



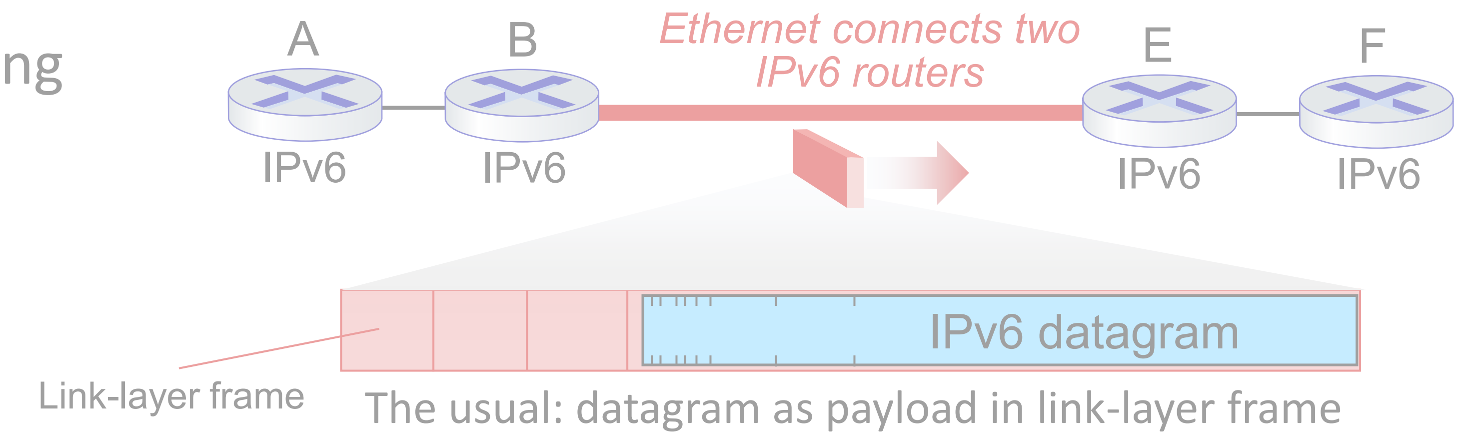
The usual: datagram as payload in link-layer frame

IPv4 network connecting two IPv6 routers

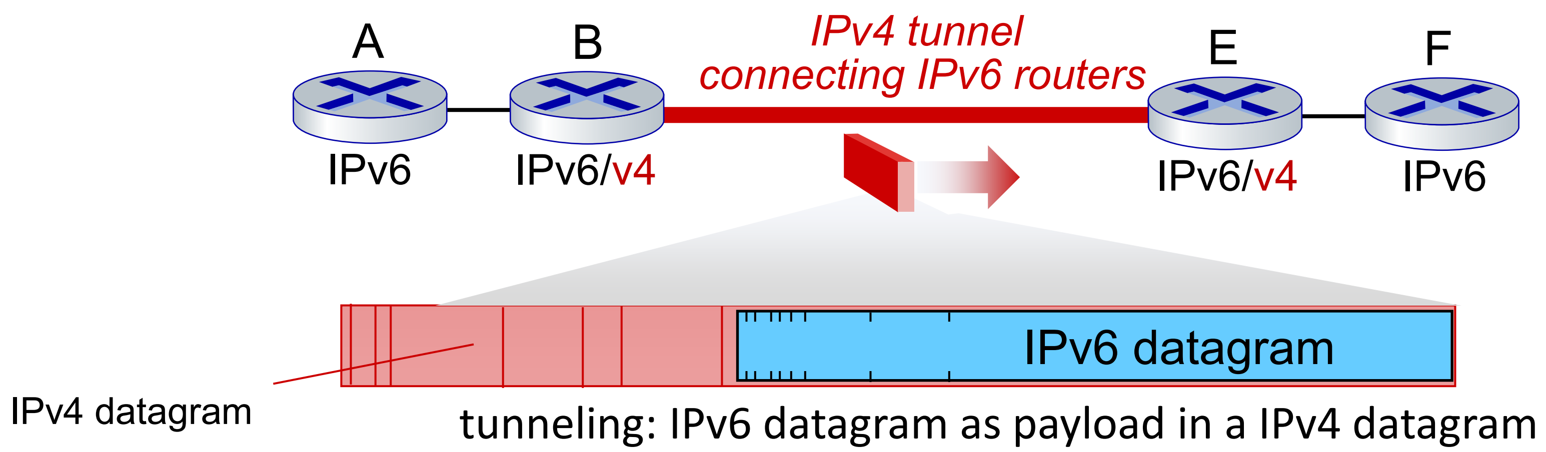


Tunneling and encapsulation

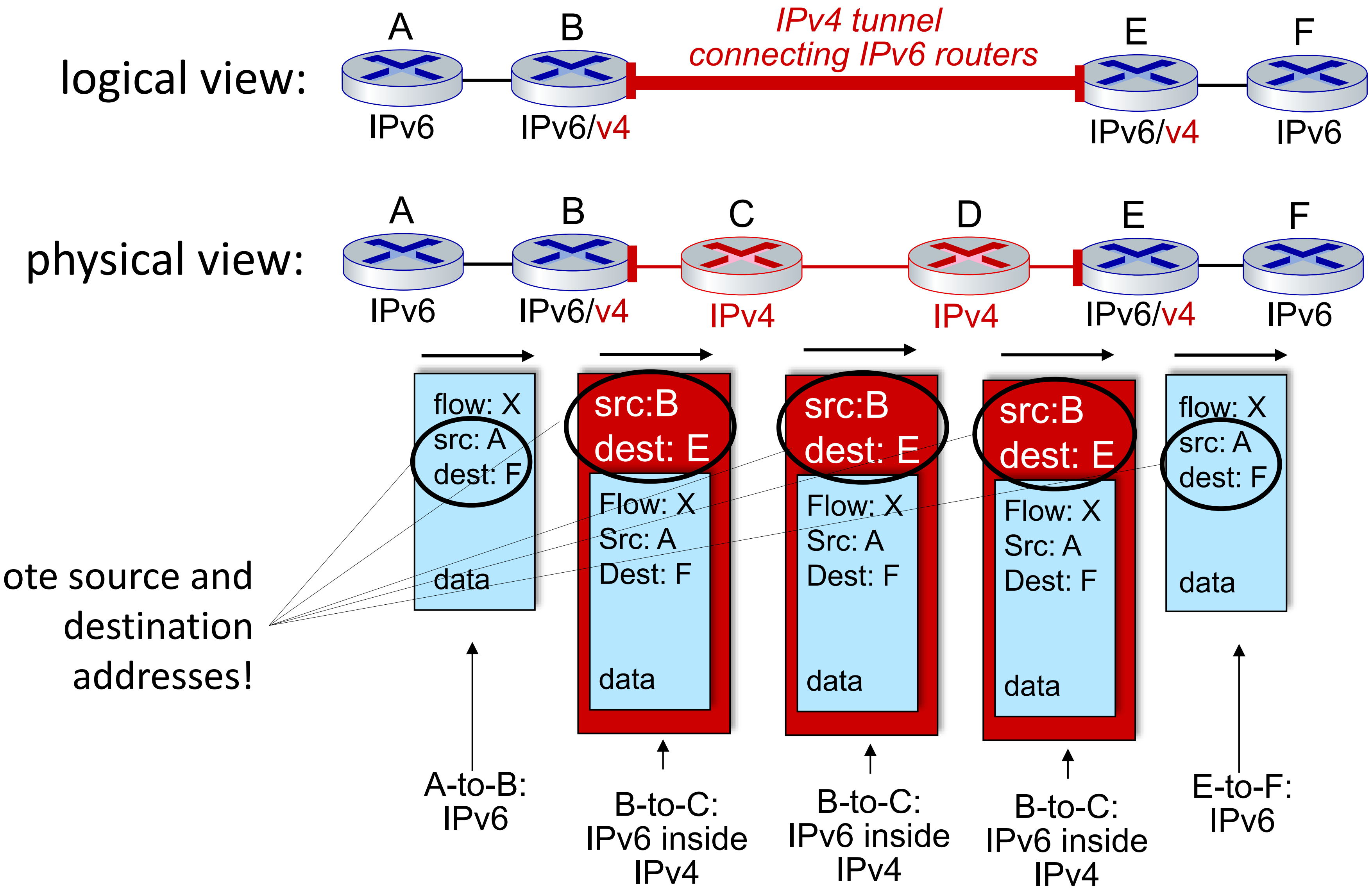
Ethernet connecting two IPv6 routers:

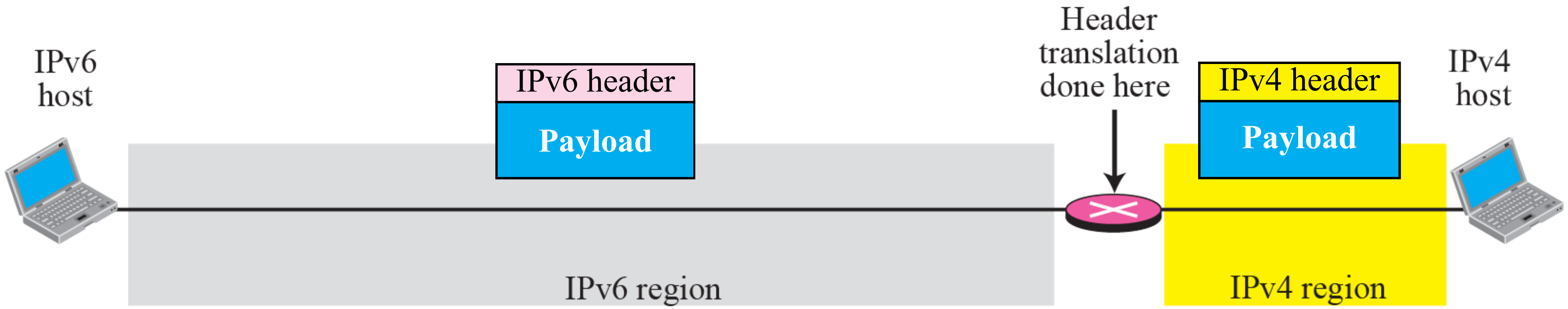


IPv4 tunnel connecting two IPv6 routers



Tunneling


A
B
IPv4 tunnel connecting IPv6 routers
E
F
logical view:
IPv6
IPv6/v4
IPv6/v4
IPv6
A
B
C
D
E
F
physical view:
IPv6
IPv6/v4
IPv4
IPv4
IPv6/v4
IPv6
flow: X
src: A
dest: F
data
src:B
dest: E
Flow: X
Src: A
Dest: F
data
src:B
dest: E
Flow: X
Src: A
Dest: F
data
src:B
dest: E
Flow: X
Src: A
Dest: F
data
flow: X
src: A
dest: F
data
Note source and destination addresses!
A-to-B: IPv6
B-to-C: IPv6 inside IPv4
B-to-C: IPv6 inside IPv4
B-to-C: IPv6 inside IPv4
E-to-F: IPv6

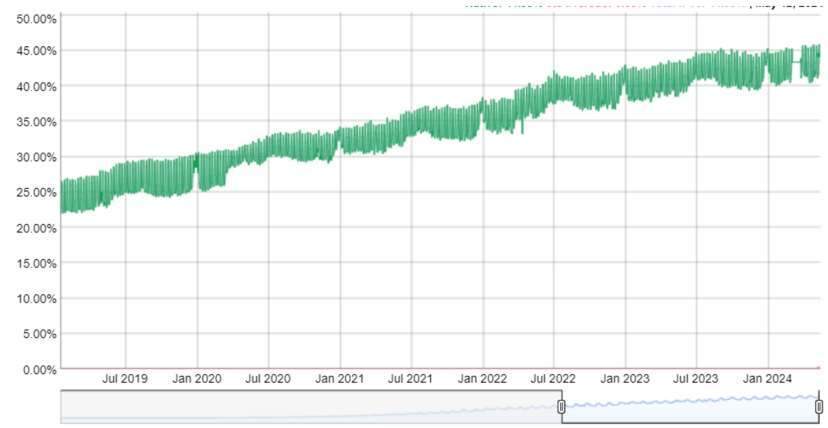


Network Address Translation - Protocol Translation (NAT-PT) [RFC 2766]

- With the help of the NAT Protocol Translation technique, the IPv4 and IPv6 networks can also communicate with each other, which do not understand the address of different IP versions.
- It removes the sender's IP version header and adds the receiver's IP version header.
- DNS64+NAT64 in sender side IPv6 network to communicate with IPv4 receiver network.
- 64:ff9b::/96 is called a well-known prefix for ipv4 to ipv6 translation. DNS64 converts ipv4 to ipv6 and NAT64 converts in IPv6 to ipv4

IPv6: adoption

- Google[1]: ~ 45% of clients access services via IPv6
- India is leading in IPv6 expansion.
- Long (long!) time for deployment, use
  - 25 years and counting!
  - think of application-level changes in last 25 years: WWW, social media, streaming media, gaming, telepresence, …
  - *Why?*



IPv6 connectivity among Google users statistics (data by Google, May 2024)

https://www.ipxo.com/blog/ipv6-adoption/